

整数集 $\mathbb{Z}$ 上的运算

定义 ($\mathbb{Z}$ 上的加法逆元运算) 对 $\forall [m, n] \in \mathbb{Z}$, 定义 $[m, n]$ 的加法逆元为: $-[m, n] := [n, m]$

Lemma: $\mathbb{Z}$ 上的加法逆元运算良定义.

Proof: 对 $\forall [m, n], [m', n'] \in \mathbb{Z}$. 若 $[m, n] = [m', n']$, 则有:

$$\therefore [m, n] = [m', n'] \quad \therefore (m, n) \sim (m', n') \quad \therefore m + n' = m' + n$$

$$\therefore n + m' = n' + m \quad \therefore (n, m) \sim (n', m') \quad \therefore [n, m] = [n', m']$$

$$\therefore -[m, n] = [n, m] = [n', m'] = -[m', n'] \quad \square$$

Lemma ($\mathbb{Z}$ 中的加法逆元性质) 对 $\forall x \in \mathbb{Z}$, 有: $x + (-x) = 0 = (-x) + x$.

Proof: 对 $\forall [m, n] \in \mathbb{Z}$, 有:

$$[m, n] + (-[m, n]) = [m, n] + [n, m] = [m+n, n+m] = [0, 0]$$

$$(-[m, n]) + [m, n] = [n, m] + [m, n] = [n+m, m+n] = [0, 0]$$

$$\therefore [m, n] + (-[m, n]) = [0, 0] = (-[m, n]) + [m, n] \quad \square$$

Lemma ($\mathbb{Z}$ 中加法逆元的唯一性) 对 $\forall x \in \mathbb{Z}$, 存在唯一的 $-x \in \mathbb{Z}$, 使得

$$x + (-x) = 0$$

Proof: 存在性上面的引理已证. 下证唯一性.

假设 $\exists \alpha, \beta \in \mathbb{Z}$, s.t. $x + \alpha = 0$ 且 $x + \beta = 0$

$$\therefore x + \alpha = 0 = x + \beta$$

$$\therefore \alpha + x = x + \alpha = x + \beta = \beta + x$$

$$\therefore \alpha, \beta, x \in \mathbb{Z}, \alpha + x = \beta + x \quad \therefore \alpha = \beta \quad \therefore \text{唯一性得证} \quad \square$$

Lemma: 对 $\forall x \in \mathbb{Z}$, 有: $-(-x) = x$

Proof: $\because x \in \mathbb{Z} \quad \therefore -x \in \mathbb{Z}$

$$\therefore x + (-x) = 0 \quad \therefore (-x) + x = 0$$

$$\therefore (-x) + (-(-x)) = 0$$

$$\therefore \text{由上一引理中的唯一性可得: } -(-x) = x \quad \square$$

定义 ($\mathbb{Z}$ 中的减法) 对 $\forall x, y \in \mathbb{Z}$, 定义 $x - y$ 为:

$$x - y := x + (-y)$$

Lemma: 对 $\forall x \in \mathbb{Z}$, 有: $(-1) \cdot x = -x$

Proof: $\because 1 \in \mathbb{Z}$ 是整数集 $\mathbb{Z}$ 的乘法幺元 $\therefore -1 \in \mathbb{Z}$

$$\therefore x + (-1) \cdot x = 1 \cdot x + (-1) \cdot x = (1 + (-1)) \cdot x = 0 \cdot x = 0$$

$$\text{又: } x + (-x) = 0$$

$$\therefore \text{由加法逆元的唯一性, 得: } (-1) \cdot x = -x \quad \square$$

Lemma: 对 $\forall x, y \in \mathbb{Z}$, 有: $(-x) + (-y) = -(x+y)$

Proof: $\because x \in \mathbb{Z} \quad \therefore \exists$ 唯一的 $-x \in \mathbb{Z}$, s.t. $x + (-x) = 0$.

$\because y \in \mathbb{Z} \quad \therefore \exists$ 唯一的 $-y \in \mathbb{Z}$, s.t. $y + (-y) = 0$.

$\therefore x+y \in \mathbb{Z}$, 且有:

$$\begin{aligned}(x+y) + ((-x) + (-y)) &= ((x+y) + (-x)) + (-y) \\&= ((y+x) + (-x)) + (-y) = (y + (x + (-x))) + (-y) \\&= (y+0) + (-y) = y + (-y) = 0\end{aligned}$$

$$\therefore (-x) + (-y) = -(x+y) \quad \square$$

定理: $(\mathbb{Z}, +, \cdot, 0, 1)$ 是一个交换环.

Proof: 所有关于交换环的条件, 都已证明. $\square$

Lemma: 对 $\forall x, y, z \in \mathbb{Z}$. 若 $x \geq y$, 则有: $x+z \geq y+z$

Proof: $\because x \geq y \quad \therefore y \leq x \quad \therefore x-y \in \mathbb{Z}_{\geq 0}$ ~~$\therefore x-y+0 \in \mathbb{Z}_{\geq 0}$~~

$$\therefore x-y+0 = x-y \in \mathbb{Z}_{\geq 0}$$

$\because z \in \mathbb{Z} \quad \therefore \exists$ 唯一的 $-z \in \mathbb{Z}$, s.t. $z + (-z) = 0$

$$\begin{aligned}\therefore x-y+0 &= x-y + (z+(-z)) = x+(-y) + z+(-z) = x+z + ((-y)+(-z)) \\&= x+z + (-(y+z)) = (x+z) - (y+z)\end{aligned}$$

$$\therefore (x+z) - (y+z) = x-y+0 \in \mathbb{Z}_{\geq 0} \quad \therefore y+z \leq x+z \quad \therefore x+z \geq y+z \quad \square$$

Lemma: $\forall x \in \mathbb{Z}$, 有: $x \geq 0 \Leftrightarrow -x \leq 0$

Proof: $\forall x \in \mathbb{Z}$, $\exists$ 唯一的 $-x \in \mathbb{Z}$, s.t. $x + (-x) = 0$.

($\Rightarrow$): $\because x \geq 0, -x \in \mathbb{Z} \quad \therefore x + (-x) \geq 0 + (-x)$

$$\therefore 0 \geq -x \quad \therefore -x \leq 0$$

($\Leftarrow$): $\because -x \leq 0 \quad \therefore 0 \geq -x$

$$\therefore 0 \geq -x, x \in \mathbb{Z}$$

$$\therefore 0 + x \geq (-x) + x \quad \therefore x \geq 0 \quad \square$$

Lemma: $\forall x, y \in \mathbb{Z}$, 有: $-(x-y) = y-x$

Proof: $\because (x-y) + (y-x) = (x+(-y)) + (y+(-x)) = x+(-y)+y+(-x)$
 $= x+0+(-x) = x+(-x) = 0$

$$\therefore y-x = -(x-y) \quad \therefore -(x-y) = y-x \quad \square$$

~~Lemma: $\forall x, y, z \in \mathbb{Z}$. 若 $x \geq y$ 且 $z \geq 0$, 则有: $xz \geq yz$~~

Lemma: $\forall x = [m, n] \in \mathbb{Z}$, 有: $x \geq 0 \Leftrightarrow m \geq n$

Proof: ($\Rightarrow$): $\because x \geq 0 \quad \therefore 0 \leq x \quad \therefore x - 0 \in \mathbb{Z}_{\geq 0}$

$$\because x - 0 = [m, n] - [0, 0] = [m, n] + (-[0, 0])$$

$$= [m, n] + [0, 0] = [m+0, n+0] = [m, n]$$

$$\therefore \cancel{[m, n]} \quad x = [m, n] = x - 0 \in \mathbb{Z}_{\geq 0} \quad \therefore x = [x, 0]$$

$\downarrow$
 $\mathbb{Z}_{\geq 0}$ 嵌入 $\mathbb{Z}$.

$$\therefore [m, n] = x = [x, 0]$$

$$\therefore (m, n) \sim (x, 0) \quad \therefore m+0 = x+n$$

$$\therefore m = m+0 = x+n = n+x$$

$$\therefore m \in \mathbb{Z}_{\geq 0}, n \in \mathbb{Z}_{\geq 0}, x \in \mathbb{Z}_{\geq 0}, m = n+x \quad \therefore n \leq m$$

$$\therefore m \geq n.$$

$$(\Leftarrow): \therefore m \in \mathbb{Z}_{\geq 0}, n \in \mathbb{Z}_{\geq 0}, m \geq n$$

$$\therefore m \in \mathbb{Z}_{\geq 0}, n \in \mathbb{Z}_{\geq 0}, n \leq m$$

$$\therefore \exists \alpha \in \mathbb{Z}_{\geq 0}, \text{ s.t. } m = n + \alpha$$

$$\therefore m+0 = m = n+\alpha = \alpha+n \quad \therefore (m, n) \sim (\alpha, 0)$$

$$\therefore [m, n] = [\alpha, 0] = \alpha \quad \therefore x = [m, n] = \alpha.$$

$$\therefore 0+0=0 \quad \therefore -0=0$$

$$\therefore x-0 = x+(-0) = x+0 = x = \alpha \in \mathbb{Z}_{\geq 0}$$

$$\therefore 0 \leq x \quad \therefore x \geq 0$$



~~Lemma: 对 $\forall x = [m, n] \in \mathbb{Z}$, 有: $x \leq 0 \Leftrightarrow m \leq n$~~

Lemma: 对 $\forall x \in \mathbb{Z}$, 有: $x \leq 0 \Leftrightarrow -x \geq 0$

Proof: 对 $\forall x \in \mathbb{Z}$, $\exists$ 唯一的 $-x \in \mathbb{Z}$, s.t. $x+(-x)=0$

$$(\Rightarrow): \therefore x \leq 0 \quad \therefore 0 \geq x \quad \therefore 0+(-x) \geq x+(-x) \quad \therefore -x \geq 0$$

$$(\Leftarrow): \therefore -x \geq 0 \quad \therefore (-x)+x \geq 0+x \quad \therefore 0 \geq x \quad \therefore x \leq 0$$



Lemma: $\forall x = \llbracket m, n \rrbracket \in \mathbb{Z}$, 有: $x \leq 0 \Leftrightarrow m \leq n$

Proof: $x \leq 0 \Leftrightarrow -x \geq 0$
 $\Leftrightarrow -\llbracket m, n \rrbracket \geq 0$
 $\Leftrightarrow \llbracket n, m \rrbracket \geq 0$
 $\Leftrightarrow n \geq m$
 $\Leftrightarrow m \leq n$ □

Lemma: $\forall x, y \in \mathbb{Z}$, 有: $x \geq y \Leftrightarrow x - y \geq 0$

Proof: $(\Rightarrow)$: $\because x \geq y \quad \therefore y \leq x \quad \therefore x - y \in \mathbb{Z}_{\geq 0}$
 $\therefore$ 由 A.1 — Peano 算术 / 5 笔记知: $x - y \geq 0$
 $(\Leftarrow)$: $\because x, y \in \mathbb{Z} \quad \therefore x - y \in \mathbb{Z}$
 $\therefore x - y \geq 0 \quad \therefore 0 \leq x - y \quad \therefore (x - y) - 0 \in \mathbb{Z}_{\geq 0}$
 $\therefore (x - y) - 0 = (x - y) + (-0) = (x - y) + 0 = x - y$
 $\therefore x - y = (x - y) - 0 \in \mathbb{Z}_{\geq 0} \quad \therefore y \leq x \quad \therefore x \geq y$ □

Lemma: $\forall x \in \mathbb{Z}$, 有: $x \geq 0 \Leftrightarrow x \in \mathbb{Z}_{\geq 0}$

Proof: $(\Leftarrow)$: A.1 — Peano 算术 / 5 笔记已证.
 $(\Rightarrow)$: $\because x \geq 0 \quad \therefore 0 \leq x \quad \therefore x - 0 \in \mathbb{Z}_{\geq 0}$
 $\therefore x - 0 = x + (-0) = x + 0 = x$
 $\therefore x = x - 0 \in \mathbb{Z}_{\geq 0}$ □

Lemma: $\forall x, y \in \mathbb{Z}$, 若 $x \geq 0$ 且 $y \geq 0$, 则有: $xy \geq 0$

Proof: $\because x, y \in \mathbb{Z} \quad \therefore \exists m, n, r, s \in \mathbb{Z}_{\geq 0}$, s.t. $x = \llbracket m, n \rrbracket, y = \llbracket r, s \rrbracket$

$$\because x \geq 0 \quad \therefore m \geq n \quad \because y \geq 0 \quad \therefore r \geq s$$

$$\because r \geq s \quad \therefore r - s \geq 0 \quad \therefore r - s \in \mathbb{Z}_{\geq 0}$$

$$\because m \in \mathbb{Z}_{\geq 0}, n \in \mathbb{Z}_{\geq 0}, r - s \in \mathbb{Z}_{\geq 0}, m \geq n$$

$$\therefore m(r - s) \geq n(r - s) \quad \therefore mr - ms \geq nr - ns$$

$$\therefore mr - ms + ms + ns \geq nr - ns + ms + ns$$

$$\therefore mr + ns \geq nr + ms$$

$$\therefore \llbracket mr + ns, nr + ms \rrbracket \geq 0 \quad \therefore \llbracket m, n \rrbracket \cdot \llbracket r, s \rrbracket \geq 0$$

$$\therefore xy \geq 0 \quad \square$$

Lemma: $\forall x, y, z \in \mathbb{Z}$. 若 $x \geq y$ 且 $z \geq 0$, 则有: $xz \geq yz$

$$\text{Proof: } \because x \geq y \quad \therefore x - y \geq 0$$

$$\because x - y \geq 0 \text{ 且 } z \geq 0 \quad \therefore (x - y)z \geq 0 \quad \therefore xz - yz \geq 0$$

$$\therefore xz \geq yz \quad \square$$

定义 ($\mathbb{Z}$ 上的序结构) 对 $\forall x, y \in \mathbb{Z}$, 定义:

$$x \leq y \iff y - x \in \mathbb{Z}_{\geq 0}$$

Lemma: $(\mathbb{Z}, \leq)$ 是^偏序集.

Proof: 对 $\forall x \in \mathbb{Z}$, $\because x - x = x + (-x) = 0 \in \mathbb{Z}_{\geq 0} \therefore x \leq x$

$\therefore$ 反身性成立.

对 $\forall x, y, z \in \mathbb{Z}$, 若 $x \leq y$ 且 $y \leq z$, 则有:

$$\because x \leq y \quad \therefore y - x \in \mathbb{Z}_{\geq 0} \quad \therefore y + (-x) \in \mathbb{Z}_{\geq 0}$$

$$\because y \leq z \quad \therefore z - y \in \mathbb{Z}_{\geq 0} \quad \therefore z + (-y) \in \mathbb{Z}_{\geq 0}$$

$$\therefore z + (-y) + y + (-x) \in \mathbb{Z}_{\geq 0} \quad \therefore z - x \in \mathbb{Z}_{\geq 0} \quad \therefore x \leq z$$

$\therefore$ 传递性成立.

对 $\forall x, y \in \mathbb{Z}$, 若 $x \leq y$ 且 $y \leq x$, 则有:

~~$$\because x \leq y \quad \therefore y - x \in \mathbb{Z}_{\geq 0} \quad \therefore y + (-x) \in \mathbb{Z}_{\geq 0}$$~~

~~$$\because y \leq x \quad \therefore x - y \in \mathbb{Z}_{\geq 0} \quad \therefore x + (-y) \in \mathbb{Z}_{\geq 0}$$~~

$$\because x \leq y \quad \therefore y \geq x \quad \therefore y - x \geq 0 \quad \therefore -(y - x) \leq 0 \quad \therefore x - y \leq 0$$

$$\because y \leq x \quad \therefore x \geq y \quad \therefore x - y \geq 0$$

$$\because x - y \leq 0 \text{ 且 } x - y \geq 0 \quad \therefore x - y = 0 \quad \therefore x = y \quad \therefore \text{对称性成立.}$$

$\therefore (\mathbb{Z}, \leq)$ 是偏序集.

